

Electromagnetism & Electrodynamics HW6

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Technion

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1 Planes at an angle

1.1 Exactly in the middle

To solve this using the images method I'll add image charges using the following logic: for any charge that's "up" relative to each of the planes, regardless of whether it's real or an image, I'll add an image charge of it "below" the plane. In total there'll be 5 image charges. The first image charge is at a mirrored y position relative to the original charge. The 2nd charge is mirrored with regard to the plane $y = \sqrt{3}x$ which can be viewed as a straight line going out at the angle $\frac{\pi}{3}$ relative to the x axis. Therefore the 2nd image charge will have an angle of $\frac{\pi}{2}$, or in other words, along the y axis. the distance from the origin must remain the same as the original charge. We then just mirror the 3 charges we have at this point relative to the plane perpendicular to $y = x/\sqrt{3}$ with opposite charge sign and get the rest of the charges. Since the charge is along that line, and $\sqrt{x^2 + y^2} = d$ we also find that for the original charge $x = \frac{\sqrt{3}}{2}d$ and $y = \frac{1}{2}d$

$$e_0 = e \qquad r_0 = \begin{pmatrix} \frac{\sqrt{3}}{2}d \\ \frac{1}{2}d \\ 0 \end{pmatrix} \qquad (1)$$

$$e_1 = -e \qquad r_1 = \begin{pmatrix} \frac{\sqrt{3}}{2}d \\ -\frac{1}{2}d \\ 0 \end{pmatrix} \qquad (2)$$

$$e_2 = -e \qquad r_2 = \begin{pmatrix} 0 \\ d \\ 0 \end{pmatrix} \qquad (3)$$

$$e_3 = -e \qquad e_3 = \begin{pmatrix} -\frac{\sqrt{3}}{2}d \\ -\frac{1}{2}d \\ 0 \end{pmatrix} \qquad (4)$$

$$e_4 = e \qquad r_4 = \begin{pmatrix} -\frac{\sqrt{3}}{2}d \\ \frac{1}{2}d \\ 0 \end{pmatrix} \qquad (5)$$

$$e_5 = e \qquad r_5 = \begin{pmatrix} 0 \\ -d \\ 0 \end{pmatrix} \qquad (6)$$

and the potential is just the sum over those:

$$\phi = \sum_{n=0}^5 \frac{e_n}{\sqrt{(x - x_n)^2 + (y - y_n)^2 + z^2}} \qquad (7)$$

1.2 And it moved!

The generalization is fairly simple: instead of assuming the angles add up to halves of the angle between the plate, we add an offset. We'll define $\theta = \sin^{-1} \left(\frac{y_0}{d} \right)$ thus:

$$e_0 = e \qquad r_0 = \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix} d \qquad (8)$$

$$e_1 = -e \qquad r_1 = \begin{pmatrix} \cos \theta \\ -\sin \theta \\ 0 \end{pmatrix} d \qquad (9)$$

$$e_2 = -e \qquad r_2 = \begin{pmatrix} \cos \left(\frac{2\pi}{3} - \theta \right) \\ \sin \left(\frac{2\pi}{3} - \theta \right) \\ 0 \end{pmatrix} d \qquad (10)$$

$$e_3 = -e \qquad r_3 = \begin{pmatrix} \cos \left(\frac{4\pi}{3} - \theta \right) \\ \sin \left(\frac{4\pi}{3} - \theta \right) \\ 0 \end{pmatrix} d \qquad (11)$$

$$e_4 = e \qquad r_4 = \begin{pmatrix} \cos \left(\frac{2\pi}{3} + \theta \right) \\ \sin \left(\frac{2\pi}{3} + \theta \right) \\ 0 \end{pmatrix} d \qquad (12)$$

$$e_5 = e \qquad r_5 = \begin{pmatrix} \cos \left(\frac{4\pi}{3} + \theta \right) \\ \sin \left(\frac{4\pi}{3} + \theta \right) \\ 0 \end{pmatrix} d \qquad (13)$$

$$(14)$$

And the potential is:

$$\phi = \sum_{n=0}^5 \frac{e_n}{\sqrt{(x - x_n)^2 + (y - y_n)^2 + z^2}} \qquad (15)$$

1.3 Green's Function

The Green's function can be found by just taking the potential in part 2 and removing the charge itself.

$$G = \sum_{n=0}^5 \frac{e_n/e}{\sqrt{(x - x_n)^2 + (y - y_n)^2 + z^2}} \qquad (16)$$

where e_n, x_n, y_n are defined as in part 2 of this question.

2 Plane with bump

2.1 Green's Function

using the hint we construct image charges for the problem. Since for the construction of the Green's function it doesn't matter if the plane is held at an arbitrary potential or zero, it'll be easier to construct it with zero. As for the plane we can place two charges, one above it and one below it, with opposite signs. We'll place them so they're above and below and sphere as well. We then place image charges to fix the spheres using the image charge rules for spheres. This will create 2 more charges, inside the sphere. In total this gives the charges

Image	position	charge
above plane	$\mathbf{r}'$	1
below plane	$\tilde{\mathbf{r}}'$	-1
sphere above plane	$\frac{a^2}{r'^2} \mathbf{r}'$	$-\frac{a}{r'}$
sphere below plane	$\frac{a^2}{r'^2} \tilde{\mathbf{r}}'$	$\frac{a}{r'}$

which gives the green's function

$$G = \frac{1}{|\mathbf{r} - \mathbf{r}'|} - \frac{1}{|\mathbf{r} - \tilde{\mathbf{r}}'|} + \frac{a}{r'} \frac{1}{|\mathbf{r} - \frac{a^2}{r'^2} \tilde{\mathbf{r}}'|} - \frac{a}{r'} \frac{1}{|\mathbf{r} - \frac{a^2}{r'^2} \mathbf{r}'|} \quad (17)$$

expressing this explicitly in spherical coordinates

$$G = \frac{1}{\sqrt{r'^2 + r^2 - 2rr' \cos \gamma}} - \frac{1}{\sqrt{\left(\frac{rr'}{a}\right)^2 + a^2 - 2rr' \cos \gamma}} - \frac{1}{\sqrt{r^2 + r'^2 + 2rr' \cos \gamma}} + \frac{1}{\sqrt{\left(\frac{rr'}{a}\right)^2 + a^2 + 2rr' \cos \gamma}} \quad (18)$$

2.2 Derivative

2.2.1 On the sphere

$$\frac{\partial G}{\partial n'} = \frac{1}{2} \left(\frac{2r' - 2r \cos \gamma}{(r'^2 + r^2 - 2rr' \cos \gamma)^{3/2}} - \frac{2\left(\frac{r}{a}\right)^2 r' - 2r \cos \gamma}{\left(\left(\frac{rr'}{a}\right)^2 + a^2 - 2rr' \cos \gamma\right)^{3/2}} \right) \quad (19)$$

$$+ \frac{1}{2} \left(\frac{2\left(\frac{r}{a}\right)^2 r' + 2r \cos \gamma}{\left(\left(\frac{rr'}{a}\right)^2 + a^2 + 2rr' \cos \gamma\right)^{3/2}} - \frac{2r' + 2r \cos \gamma}{(r'^2 + r^2 + 2rr' \cos \gamma)^{3/2}} \right) \quad (20)$$

2.2.2 on the plane

here the derivative we're looking for is

$$-\frac{\partial G}{\partial z'} \quad (21)$$

and expressing this in nspherical coordinates, where $z' = r' \cos \theta'$

$$\frac{\partial}{\partial z'} = \cos \theta' \frac{\partial}{\partial r'} - \frac{\sin \theta'}{r'} \frac{\partial}{\partial \theta'} \quad (22)$$

but since we were taking the derivative always at $z' = 0$ then $\theta = \pi/2$ so

$$\left. \frac{\partial}{\partial z'} \right|_{z'=0} = -\frac{1}{r'} \left. \frac{\partial}{\partial \theta'} \right|_{\theta'=\pi/2} \quad (23)$$

expanding

$$\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos (\varphi - \varphi') \quad (24)$$

inserting the known theta

$$\cos \gamma = \sin \theta \cos (\varphi - \varphi') \quad (25)$$

for convenience we'll define

$$\Delta \varphi = \varphi - \varphi' \quad (26)$$

so

$$\cos \gamma = \sin \theta \cos \Delta \varphi \quad (27)$$

taking the derivative

$$-\left. \frac{\partial G}{\partial z'} \right|_{z'=0} = -\frac{2r \cos \theta}{(r^2 + r'^2 - 2rr' \sin \theta \cos \Delta \varphi)^{3/2}} + \frac{2a^3 r \cos \theta}{r'^3} \frac{1}{(r^2 + \frac{a^4}{r'^2} + 2r \frac{a^2}{r'} \sin \theta \cos \Delta \varphi)^{3/2}} \quad (28)$$

2.3 held at potential

We can use the green's function and its derivatives we found to find the potential, by plugging it into the full potential using green expression. Since there are no charges, the integral over the green function itself is zero, and since the potential on the plane is zero, we're only left with the element for the potential on the sphere.

$$\Phi(r) = -\frac{1}{4\pi} \iint_{\partial V} \Phi(r') \frac{\partial G(r, r')}{\partial n'} dS \quad (29)$$

since we're looking at the potential at $x = y = 0, z > a$,

$$\mathbf{r} = (0, 0, z) \quad z > a \quad (30)$$

translated to spherical coordiantes

$$r = z \quad \theta = 0 \quad (31)$$

and

$$r' = a \quad 0 \leq \theta' \leq \pi/2 \quad (32)$$

and if we define

$$A_- = \sqrt{z^2 + a^2 - 2az \cos \theta'} \quad A_+ = \sqrt{z^2 + a^2 + 2az \cos \theta'} \quad (33)$$

and notice that

$$\cos \gamma = \cos \theta \cos \theta' + \sin \theta \sin \theta' \cos \Delta \varphi = \cos \theta' \quad (34)$$

then we have

$$\frac{\partial G}{\partial n'} = \frac{1}{2} \left(\frac{2a - 2z \cos \gamma}{(a^2 + z^2 - 2az \cos \gamma)^{3/2}} - \frac{2 \left(\frac{z}{a}\right)^2 a - 2z \cos \gamma}{\left(\left(\frac{az}{a}\right)^2 + a^2 - 2az \cos \gamma\right)^{3/2}} \right) \quad (35)$$

$$+ \frac{1}{2} \left(\frac{2 \left(\frac{z}{a}\right)^2 a + 2z \cos \gamma}{\left(\left(\frac{az}{a}\right)^2 + a^2 + 2az \cos \gamma\right)^{3/2}} - \frac{2a + 2z \cos \gamma}{(a^2 + z^2 + 2az \cos \gamma)^{3/2}} \right) \quad (36)$$

$$= \frac{a - z \cos \theta'}{(a^2 + z^2 - 2az \cos \theta')^{3/2}} - \frac{\frac{z^2}{a} - z \cos \theta'}{(z^2 + a^2 - 2az \cos \theta')^{3/2}} \quad (37)$$

$$+ \frac{\frac{z^2}{a} + z \cos \theta'}{(z^2 + a^2 + 2az \cos \theta')^{3/2}} - \frac{a + z \cos \theta'}{(a^2 + z^2 + 2az \cos \theta')^{3/2}} \quad (38)$$

$$= \frac{a - z \cos \theta'}{A_-^3} - \frac{\frac{z^2}{a} - z \cos \theta'}{A_-^3} + \frac{\frac{z^2}{a} + z \cos \theta'}{A_+^3} - \frac{a + z \cos \theta'}{A_+^3} \quad (39)$$

$$= \frac{a - \frac{z^2}{a}}{A_-^3} + \frac{\frac{z^2}{a} - a}{A_+^3} \quad (40)$$

$$= \frac{z^2 - a^2}{a} \left(\frac{1}{A_+^3} - \frac{1}{A_-^3} \right) \quad (41)$$

Therefore

$$\Phi(z) = -\frac{V_0}{4\pi} \int_0^{2\pi} d\varphi' \int_0^{\pi/2} d\theta' a^2 \sin \theta' \frac{z^2 - a^2}{a} \left(\frac{1}{A_+^3} - \frac{1}{A_-^3} \right) \quad (42)$$

$$= \frac{V_0 a (z^2 - a^2)}{2} \int_0^{\pi/2} d\theta' \sin \theta' \left(\frac{1}{A_-^3} - \frac{1}{A_+^3} \right) \quad (43)$$

$$= \frac{V_0 a (z^2 - a^2)}{2} \int_0^{\pi/2} d\theta' \sin \theta' \left(\frac{1}{\sqrt{z^2 + a^2 - 2az \cos \theta'}^3} - \frac{1}{\sqrt{z^2 + a^2 + 2az \cos \theta'}^3} \right) \quad (44)$$

$$[u = \cos \theta', du = -\sin \theta' d\theta'] = \frac{V_0 a (z^2 - a^2)}{2} \int_0^1 du \left(\frac{1}{\sqrt{z^2 + a^2 - 2azu}^3} - \frac{1}{\sqrt{z^2 + a^2 + 2azu}^3} \right) \quad (45)$$

$$= \frac{V_0 a (z^2 - a^2)}{2} \left(\frac{1}{az} \left(\frac{1}{z - a} - \frac{1}{\sqrt{z^2 + a^2}} \right) - \frac{1}{az} \left(\frac{1}{\sqrt{z^2 + a^2}} - \frac{1}{z + a} \right) \right) \quad (46)$$

$$= \frac{V_0 (z^2 - a^2)}{2z} \left(\frac{1}{z - a} + \frac{1}{z + a} - \frac{2}{\sqrt{z^2 + a^2}} \right) \quad (47)$$

$$= V_0 \left(1 - \frac{z^2 - a^2}{z\sqrt{z^2 + a^2}} \right) \quad (48)$$

3 split plane

3.1 Symmetry under inflation

First we show the boundary conditions do not change. The boundary conditions are:

$$\phi(x > 0, y = 0, z) = 0 \qquad \phi(x < 0, y = 0, z) = V \quad (49)$$

We'll perform the transformation $(x, y, z) \rightarrow \alpha(x, y, z)$ where $\alpha > 0$. Since alpha is positive, it does not change the sign of x . 0 remains 0, and z is arbitrary, therefore the boundary conditions do not change under the transformation. Next we look at the laplace equation under the same transformation

$$\nabla^2 \phi(\alpha x, \alpha y, \alpha z) = \alpha^2 \nabla^2 \phi(x, y, z) = 0 \quad (50)$$

Next we look at the problem in cylindrical coordinates and inflate them in the same manner. We know the result must remain the same therefore

$$\phi(\alpha r, \theta, \alpha z) = \phi(r, \theta, z) \quad (51)$$

This means the solution cannot depend on α , which also means we only care about the relative ratio between r and z , rather than each one's value, so

$$\phi(r, \theta, z) = \phi\left(\theta, \frac{z}{r}\right) \quad (52)$$

3.2 translation in z

The value of z does not matter for the problem because regardless of what it is, everything appears exactly identical. The value given to z is simply a label. The boundary condition does not depend on z so any translation along it changes nothing either. Finally, adding a constant to z does not change ∇ so the laplace equation remains true. This also means the potential cannot depend on z , which also means it cannot depend on the ratio between z and r , thus

$$\phi(r, \theta, z) = \phi(\theta) \quad (53)$$

3.3 Green's function

Green's function in cartesian coordiantes is given by

$$G = \frac{1}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}} + \frac{1}{\sqrt{(x-x')^2 + (y+y')^2 + (z-z')^2}} \quad (54)$$

We'll move to cylindrical coordinates using

$$x = r \cos \theta \qquad y = r \sin \theta \quad (55)$$

$$x' = r' \cos \theta' \qquad y' = r' \sin \theta' \quad (56)$$

and find (omitting the full development for brevity)

$$G = \frac{1}{\sqrt{r^2 + r'^2 - 2rr' \cos(\theta - \theta') + (z - z')^2}} - \frac{1}{\sqrt{r^2 + r'^2 - 2rr' \cos(\theta + \theta') + (z - z')^2}} \quad (57)$$

3.4 Top half

The potential is given by

$$\phi(\mathbf{x}) = -\frac{1}{4\pi} \int_{y'=0, x<0} dl \phi(\mathbf{x}') \frac{\partial G(\mathbf{x}, \mathbf{x}')}{\partial n'} \quad (58)$$

$$= -\frac{1}{4\pi} \int_{\theta'=\pi} dl \phi(\mathbf{x}') \frac{\partial G(\mathbf{x}, \mathbf{x}')}{\partial n'} \quad (59)$$

$$= \frac{1}{4\pi} \int_{\theta'=\pi} dl \phi(\mathbf{x}') \frac{\partial G(\mathbf{x}, \mathbf{x}')}{\partial y'} \quad (60)$$

$$= \frac{V}{2\pi} \int_{-\infty}^0 dx' \int_{-\infty}^{\infty} dz' \frac{y}{[(x - x')^2 + y^2 + (z - z')^2]^{3/2}} \quad (61)$$

$$= \frac{V}{2\pi} \int_{-\infty}^0 dx' \frac{y}{(x - x')^2 + y^2} \cdot \left. \frac{z' - z}{\sqrt{(x - x')^2 + y^2 + (z - z')^2}} \right|_{z=-\infty}^{\infty} \quad (62)$$

$$= \frac{V}{\pi} \int_{-\infty}^0 dx' \frac{y}{(x - x')^2 + y^2} \quad (63)$$

$$= \frac{V}{\pi} \arctan \left(\frac{x - x'}{y} \right) \Big|_{-\infty}^0 \quad (64)$$

$$= V \left(\frac{1}{2} - \frac{1}{\pi} \arctan \left(\frac{x}{y} \right) \right) \quad (65)$$

3.5 surface charge

Since there is an infinite charged plane, we know the surface charge is given by

$$\sigma(x) = \frac{1}{4\pi} [E_y(x, y = 0^+) - E_y(x, y = 0^-)] \quad (66)$$

The field above the plate at 0 is

$$E_y(x, y = 0^+) = -\frac{V}{\pi x} \quad (67)$$

Next we notice that if we reflect the y axis, everything remains the same, thus we have symmetry for flips along that axis, so

$$E(x, y = 0^-) = \frac{V}{\pi x} \quad (68)$$

thus the charge density is

$$\sigma(x) = -\frac{V}{2\pi^2 x} \quad (69)$$

3.6 A ring of charge

$$\rho(r, z) = \frac{Q}{2\pi R} \delta(r - R) \delta(z) \quad (70)$$

3.7 Another potential

The total potential is the sum of the potential as a result of the plate and the potential as a result of the charge present, i.e

$$\phi(x, y, z > 0) = \frac{V\theta}{\pi} + \iiint_V d^3x \rho(\mathbf{x}) G(\mathbf{x}, \mathbf{x}') \quad (71)$$

$$= \frac{V\theta}{\pi} + \int_0^\infty r dr \int_0^\pi d\theta \int_{-\infty}^\infty dz \frac{Q}{2\pi R} \delta(r - R) \delta(z) G(x, x') \quad (72)$$

$$= \frac{V\theta}{\pi} + \frac{Q}{2\pi} \int_0^\pi d\theta \frac{1}{\sqrt{r^2 + R^2 + z^2 - 2rR \cos(\theta - \theta')}} - \frac{1}{\sqrt{r^2 + R^2 + z^2 - 2rR \cos(\theta + \theta')}} \quad (73)$$

and as requested I'm not solving this fully.